

8 Finance

8.7 Review of Finance (Cont.)

Formulas so far:

For these formulas, we have the following: P_0 is the principal. r is the interest rate. t is the number of time intervals (specified by problem) which have passed. A is the amount paid back. I is the interest.

Simple Interest	$A = P_0(1 + r)$
Simple Interest over Time	$A = P_0(1 + rt)$

For these formulas, we have the following: P_0 is the principal, or amount of a loan. r is the annual interest rate. t is the number of years which have passed. A is the total amount. k is the number of times interest is compounded per year. d is the amount we deposit (or withdraw) from the account regularly, or the payment amount on a loan.

Compound Interest	$A = P_0 \left(1 + \frac{r}{k}\right)^{kt}$
Savings Annuity	$A = d \frac{\left(1 + \frac{r}{k}\right)^{kt} - 1}{\frac{r}{k}}$
Payout Annuity/Loans	$P_0 = d \frac{1 - \left(1 + \frac{r}{k}\right)^{-kt}}{\frac{r}{k}}$

Example 8.7.1 Marcy received an inheritance of \$20,000, and invested it at 6% interest. She is going to use it for college, withdrawing money for tuition and expenses each quarter. How much can she take out each quarter if she has 3 years of school left?

Example 8.7.2 Paul wants to buy a new car. Rather than take out a loan, he decides to save \$200 a month in an account earning 3% interest compounded monthly. How much will he have saved up after 3 years?

Example 8.7.3 Keisha is managing investments for a non-profit company. They want to invest some money in an account earning 5% interest compounded annually with the goal to have \$30,000 in the account in 6 years. How much should Keisha deposit into the account?

Example 8.7.4 Miao is going to finance new office equipment at a 2% rate over a 4 year term. If she can afford monthly payments of \$100, how much new equipment can she buy?

Example 8.7.5 How much would you need to save every month in an account earning 4% interest to have \$5,000 saved up in two years?

Example 8.7.6 A friend asks for a small loan of \$100 and agrees to pay it back with 10% interest in a week. For each week that goes by without payment, another \$10 is added to what they owe you. Assuming that they don't pay you back for four weeks, how much will they owe you?